

COURSE MATERIAL

SUBJECT	ENGINEERING GRAPHICS (ME23AES102)
UNIT	2
COURSE	B.TECH
SEMESTER	11
DEPARTMENT	MECHANICAL ENGINEERING
PREPARED BY (Faculty Name/s)	Dr. N RAJESH , Associate Professor
VERSION	1.0
PREPARED / REVISED DATE	01-06-2024

TABLE OF CONTENTS – UNIT 2

SNO	CONTENTS	PAGE
1	COURSE OBJECTIVES	1
2	PREREQUISITES	1
3	SYLLABUS	1
4	COURSE OUTCOMES	1
5	CO - PO/PSO MAPPING	1
6	LESSON PLAN	2
7	LECTURE NOTES	3
	2.1 INTRODUCTION – PROJECTION OF POINTS AND STRAIGHT LINES	3
	2.2 PROJECTION OF STRAIGHT LINES	4
	2.3 IMPORTANT QUESTIONS	8
8	ASSIGNMENTS	14
9	PART B QUESTIONS	15
10	SUPPORTIVE ONLINE CERTIFICATION COURSES	15
11	PRESCRIBED TEXT BOOKS & REFERENCE BOOKS	15

1. COURSE OBJECTIVES

The objectives of this course is to introduce and make the students

1. To enable with various concepts like dimensioning, conventions and standards related to Engineering Drawing
2. To impart knowledge on the Projection of Points, Lines and Plane surfaces
3. To improve the visualization skills for better understanding of projection of solids
4. To develop the imaginative skills required to understand Section of solids and Developments of surfaces.
5. To understand the viewing perception of a solid object in Isometric and orthographic projections.

2. PREREQUISITES

Students should have knowledge on
Basic Mathematics

3. SYLLABUS**UNIT II**

Orthographic Projections: Reference Plane, Importance of Reference Lines or Plane, Projections of a Point Situated in any One of the Four Quadrants.

Projections of Straight Lines: Projections of Straight Lines Parallel to both Reference Planes, Perpendicular to One Reference Plane and Parallel to other Reference Plane, Inclined to one Reference Plane and Parallel to the other Reference Plane. Projections of Straight Line Inclined to both the Reference Planes.

Projections of Planes: Regular Planes Perpendicular to both Reference Planes, Parallel to One Reference Plane and Inclined to the other Reference Plane; Plane Inclined to both the Reference Planes.

4. COURSE OUTCOMES

After completing the course, the student will be able to

1. Draw various engineering curves, scales. (L3)
2. Draw and Interpret orthographic projections of points, lines, planes. (L3)
3. Draw the projection of solids in various positions. (L3)
4. Draw and Explore the sections of solids and development of surfaces. (L3)
5. Draw an isometric and orthographic views of simple solids. (L3).

5. Co-PO / PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	2	2								2		2	2	2
CO2	2	2								2		2	2	2
CO3	2	2								2		2	2	2
CO4	2	2								2		2	2	2
CO5	2	2			1					2		2	2	2

LESSON PLAN

LECTURE	WEEK	TOPICS TO BE COVERED	REFERENCES
1	1	Introduction to projections	T1
2		Projections of points	T1, R1
3		Projections of points	T1, R1
4	2	Projections of straight lines	T1, R1
5		Projections of straight lines	T2, R1
6	3	Projections of Planes	T2, R1
7		Projections of Planes	T2, R2



6. LECTURE NOTES

2.1 INTRODUCTION

2.1.1 What is point?

An element which has no dimensions, it can be situated in the following positions with respect to principal planes of the projections.

1. Point situated above H.P and in front of V.P.
2. Point situated above H.P and behind V.P
3. Point situated below H.P and behind V.P.
4. Point situated below H.P and in front of V.P.
5. Point situated on H.P and in front of V.P.
6. Point situated above H.P and on V.P.
7. Point situated on H.P and behind V.P.
8. Point situated below H.P and on V.P.
9. Point situated on both H.P and V.P.

2.1.2 Conventional Representation:

1. Actual Position of a point designated by capitals i.e. A, B, C, D ...
2. Front view of a point is designated by small letters with dashes i.e. a' , b' , c' , d'
3. Top view of a point is designated by only small letters i.e. a , b , c , d
4. Side view of a point is designated by small letters with double dashes i.e. a'' , b'' , c'' , d'' ...

The Intersection of reference planes is a line known as reference line denoted by $x-y$ and the line connecting the front and top view is known as projection line; it is always perpendicular to the principal axis ($x-y$ line).

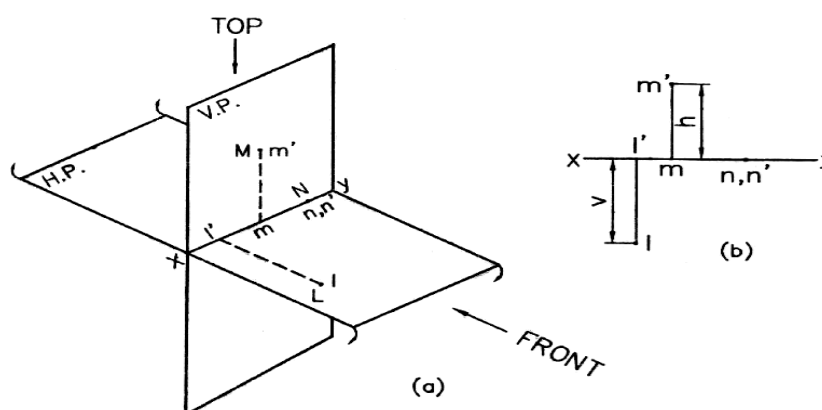


Fig.2.1

Problem:

1. Draw the orthographic projections of the following points?
 - (a) Point P is 30 mm. above H.P and 40 mm. in front of VP.
 - (b) Point Q is 25 mm. above H.P and 35 mm. behind VP.
 - (c) Point R is 32 mm. below H.P and 45 mm behind VP.
 - (d) Point S is 35 mm. below H.P and 42 mm in front of VP.
 - (e) Point T is in H.P and 30 mm behind VP.
 - (f) Point U is in V.P and 40 mm. below HP.
 - (g) Point V is in V.P and 35 mm. above H.P.
 - (h) Point W is in H.P and 48 mm. in front of VP.

Solution:

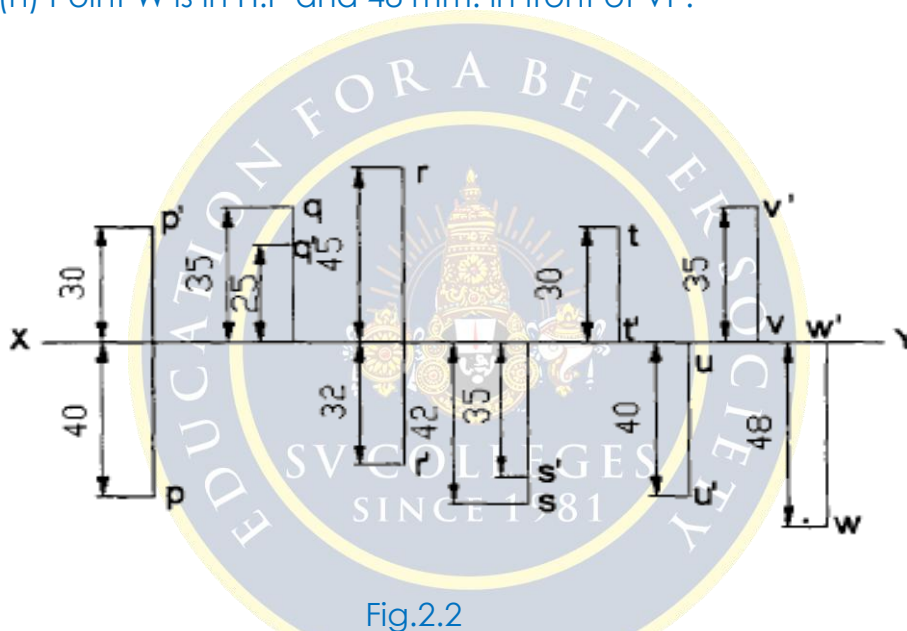


Fig.2.2

2.2 PROJECTION OF STRAIGHT LINES

2.2.1 Introduction:

What is Line?

A Shortest distance between two points and the actual length of the line is known as True Length denoted by TL.

Orientation of Straight Lines

1. Line parallel to both H.P and V.P
2. Line perpendicular to H.P and parallel to V.P
3. Line perpendicular to V.P and parallel to H.P
4. Line inclined to H.P and parallel to V.P
5. Line inclined to V.P and parallel to H.P

6. Line situated in H.P
7. Line situated in V.P
8. Line situated in both H.P and V.P
9. Line inclined to both the reference planes.
 - a. Line inclined to both H.P and V.P front view angle and top view angle = 90° .
 - b. Line inclined to both H.P and V.P front view angle and top view angle = 90° .

Problems

1. **Line parallel to both H.P and V.P** A 50mm long line AB is parallel to both H.P and V.P. The line is 25mm in front of V.P and 60mm above H.P, draw the projections of the line.

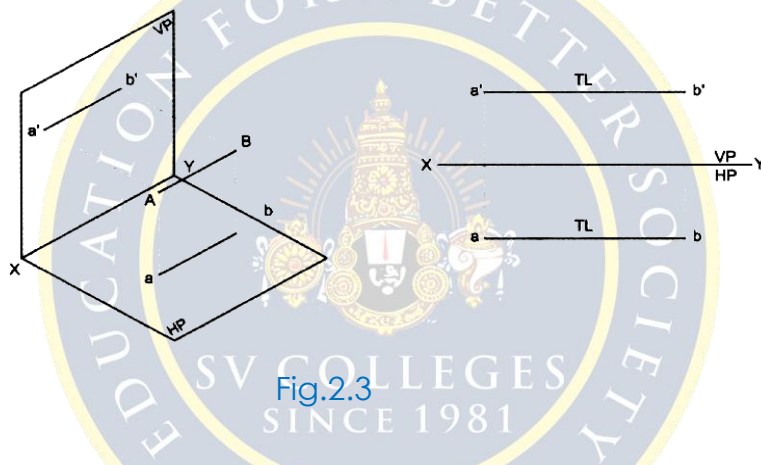


Fig.2.3

2. **Line perpendicular to H.P** A 60mm long line AB has its end A at 20mm above the H.P. The line is perpendicular to the H.P and 40mm in front of V.P, draw the projections of the line.

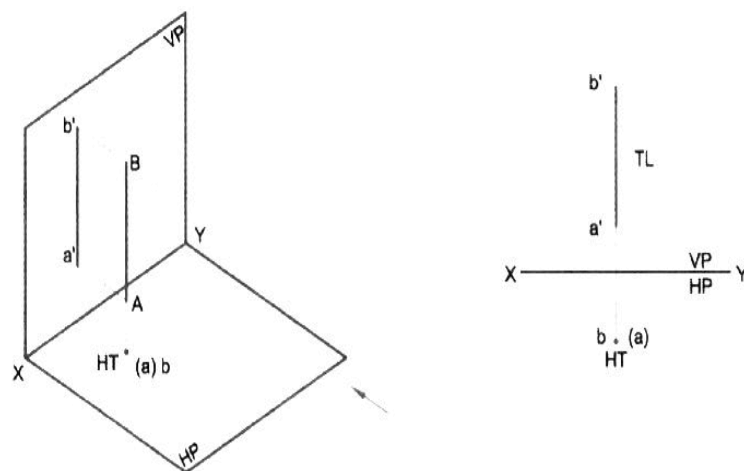


Fig.2.4

3. Line perpendicular to V.P A 60mm long line AB, has its end A at 20mm in front of the V.P. the line is perpendicular to V.P and 40mm above H.P, draw the projection of the line.

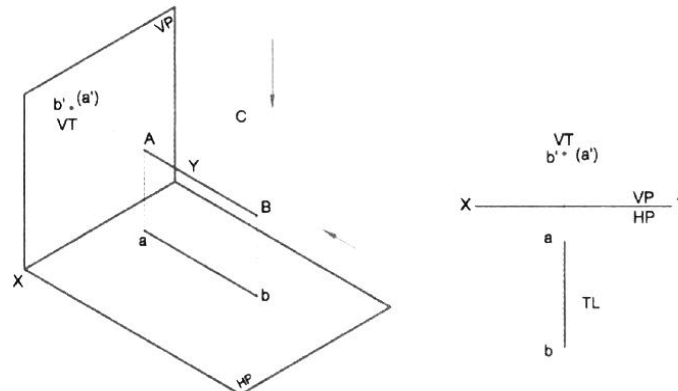


Fig.2.5

4. Line inclined to H.P and parallel to V.P A 80mm long line AB has the end A at 20mm above HP and 40mm in front of V.P. The line is inclined at 30 deg to H.P and parallel to V.P, draw the projection of the line.

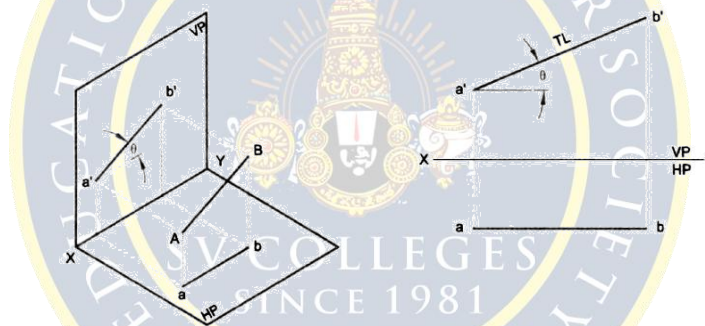


Fig.2.6

5. Line inclined to V.P and parallel to H.P An 80mm long line AB is inclined at 30 deg to V.P and is parallel to H.P. The end A is 20mm above the H.P and 20mm in front of the V.P, draw the projection of the line.

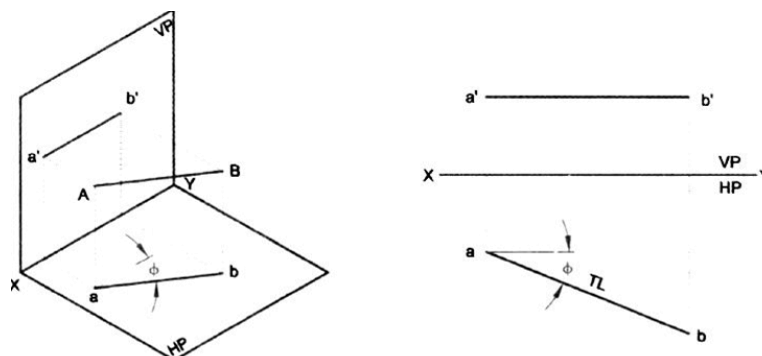


Fig.2.7

6. Line situated in H.P A line AB 60mm long is situated in H.P and inclined to V.P at 30 deg. The end A is 20mm in front of V.P, draw the projection of line.

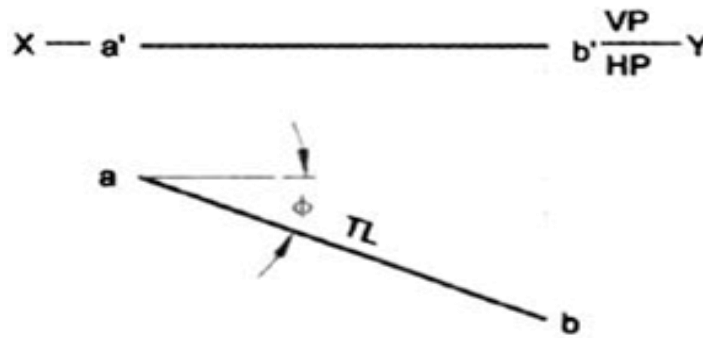


Fig.2.8

7. Line situated in V.P Draw the projections of 70mm long line AB situated in the V.P and inclined at 30 deg to H.P. The end A is 25 mm above H.P.

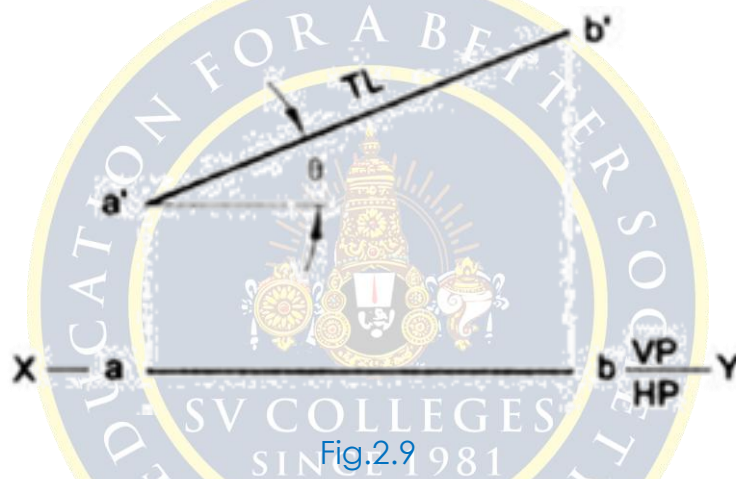


Fig.2.9

8. Lines inclined to both the reference planes. A 70mm long line AB has an end A at 20mm above H.P and 30mm in front of V.P. The line is inclined at 45 deg to the H.P and 30 deg to V.P, draw the projections.

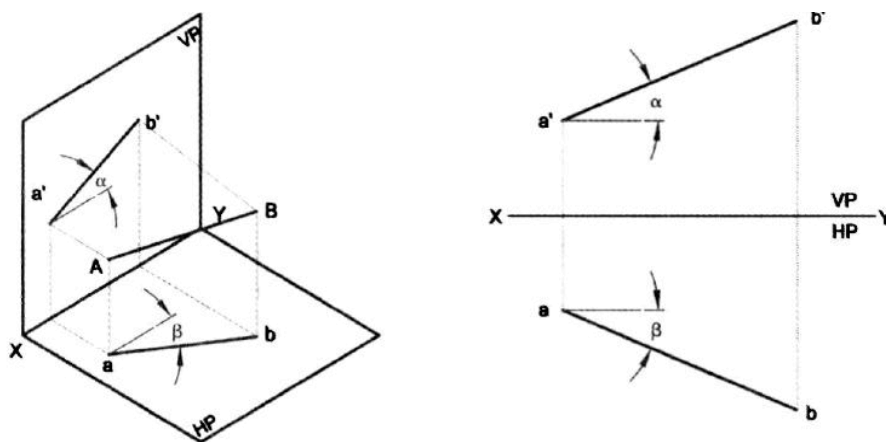


Fig.2.10

2.3 Important Questions and Previous Paper Problems:

1. A line AB, 70mm long, has its end A 15mm above HP and 20mm in front of VP. It is inclined at 30° to HP and 45° to VP. Draw its projections and mark its traces.

Solution:

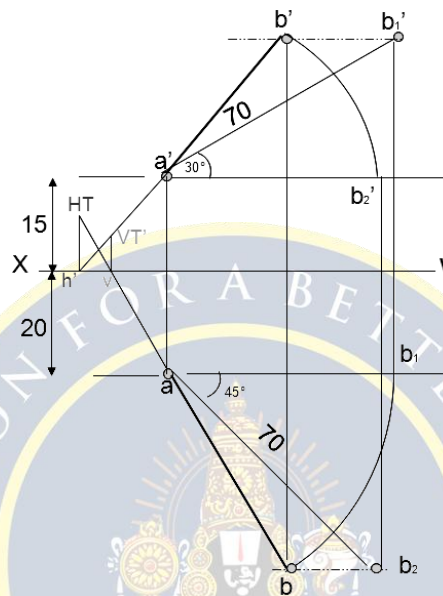


Fig.2.11

2. The top view of a 75mm long line AB measures 65mm, while its front view measures 50mm. It is one end A is in HP and 12mm in front of VP. Draw the projections of AB and determine its inclination with HP and VP

Solution:

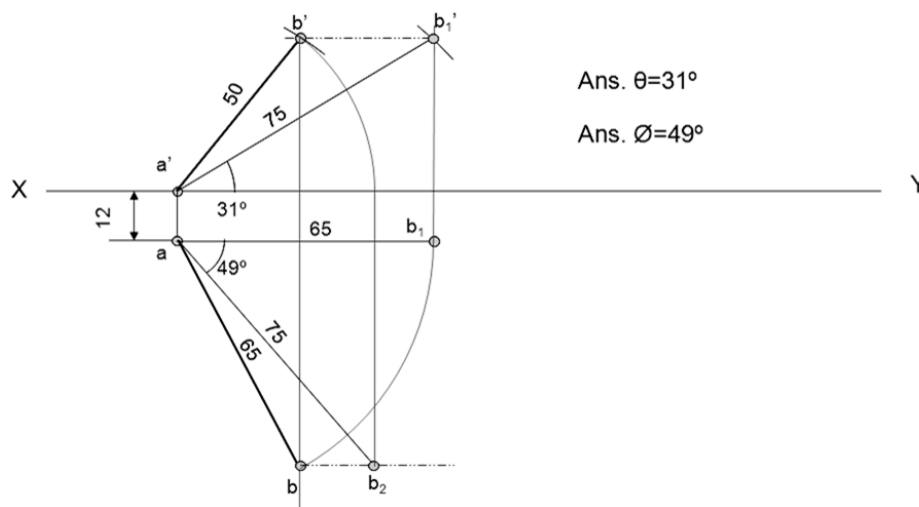
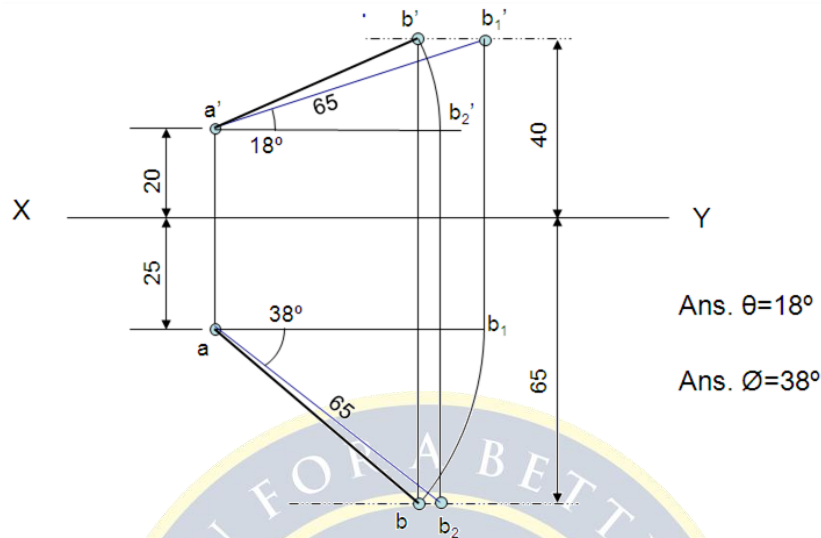


Fig.2.12

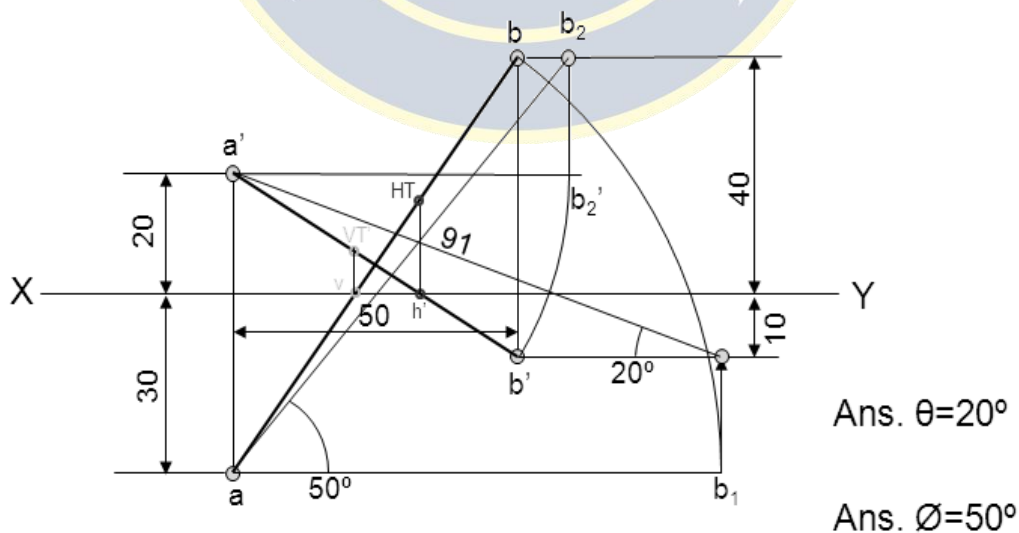
3. A line AB, 65mm long has its end A 20mm above H.P. and 25mm in front of V.P. The end B is 40mm above H.P. and 65mm in front of V.P. Draw the projections of AB and shows its inclination with H.P.

Solution:



4. The projectors of the ends of a line AB are 5cm apart. The end A is 2cm above the H.P and 3cm in front of V.P. The end B is 1cm below H.P. and 4cm behind the V.P. Determine the true length and traces of AB, and its inclination with the two planes.

Solution:



5. A line AB, 90mm long, is inclined at 45° to the H.P. and its top view makes an angle of 60° with the V.P. The end A is in the H.P. and 12mm in front of V.P. Draw its front view and find its true inclination with the V.P.

Solution:

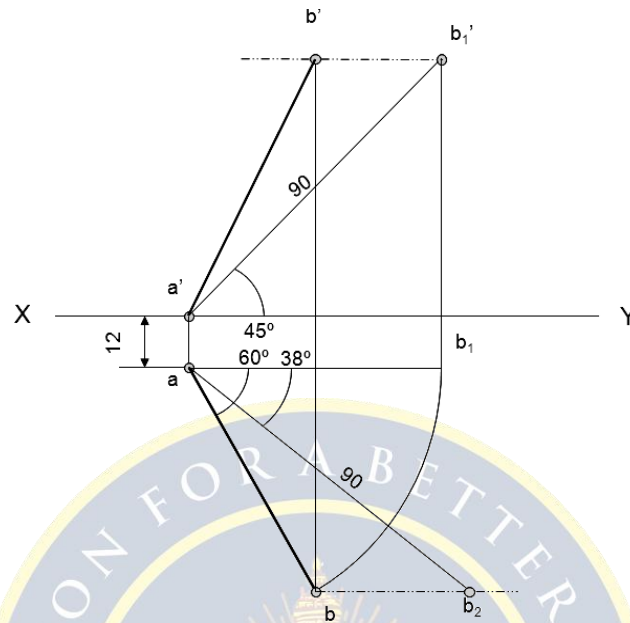


Fig.2.15

6. A line AB, 90mm long, is inclined at 30° to the H.P. Its end A is 12mm above the HP and 20mm in front of the VP. Its FV measures 65mm. Draw the TV of AB and determine its inclination with the VP.

Solution:

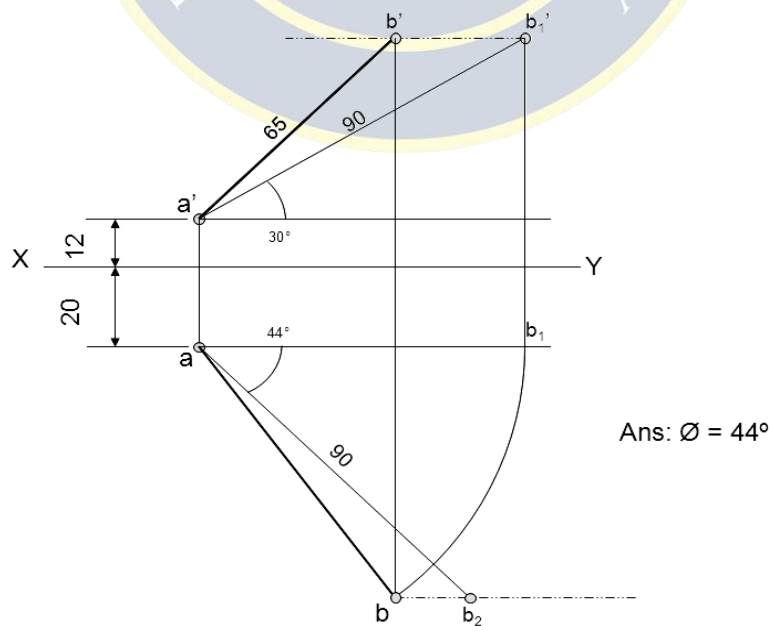


Fig.2.16

7. Two lines AB & AC make an angle of 120 deg between them in their FV & TV. AB is parallel to both the HP & VP. Determine the real angle between AB & AC.

Solution:

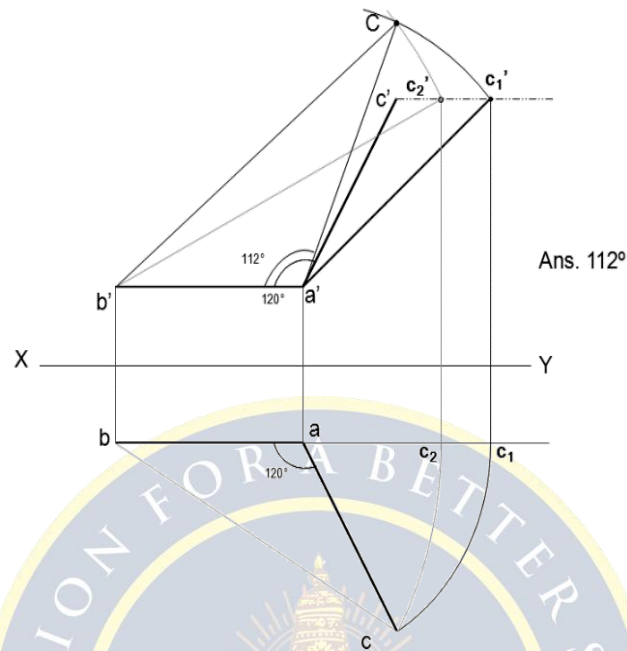


Fig.2.17

8. A line AB, inclined at 40° to the V.P. has its end 50mm and 20mm above the H.P. the length of its front view is 65mm and its V.T. is 10mm above the H.P. determine the true length of AB its inclination with the H.P. and its H.T.

Solution:

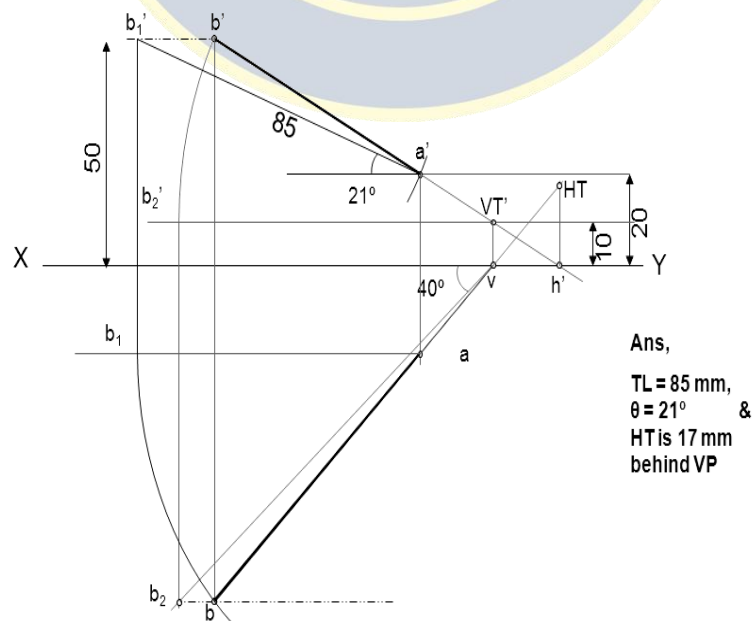


Fig.2.18

9. The top view of a 75mm long line CD measures 50 mm. C is 50 mm in front of the VP & 15mm below the HP. D is 15 mm in front of the VP & is above the HP. Draw the FV of CD & find its inclinations with the HP and the VP. Also show its traces.

Solution:

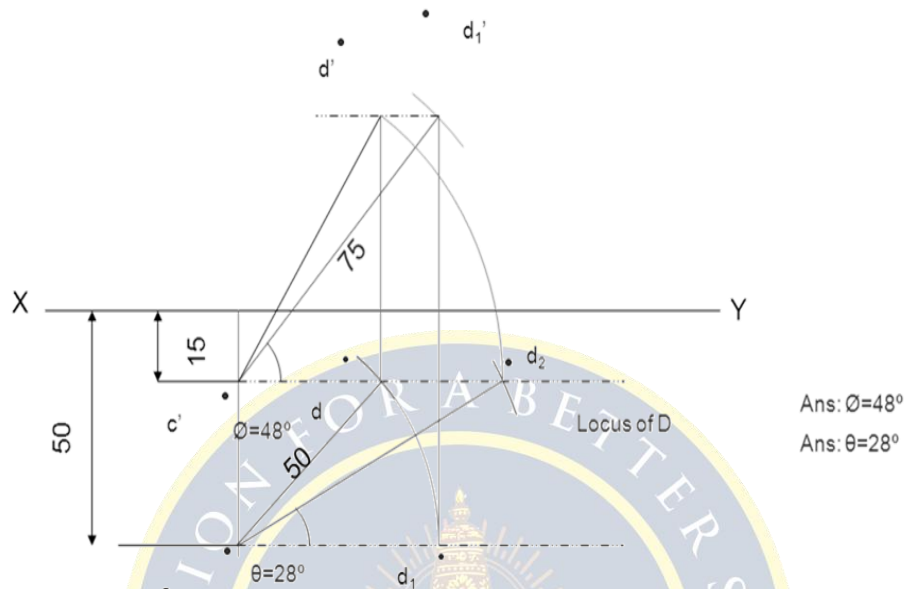


Fig.2.19

10. A line PQ 100 mm long is inclined at 30° to the H.P. and at 45° to the V.P. Its mid-point is in the V.P. and 20 mm above the H.P. Draw its projections if its end P is in the third quadrant and Q is in the first quadrant.

Solution:

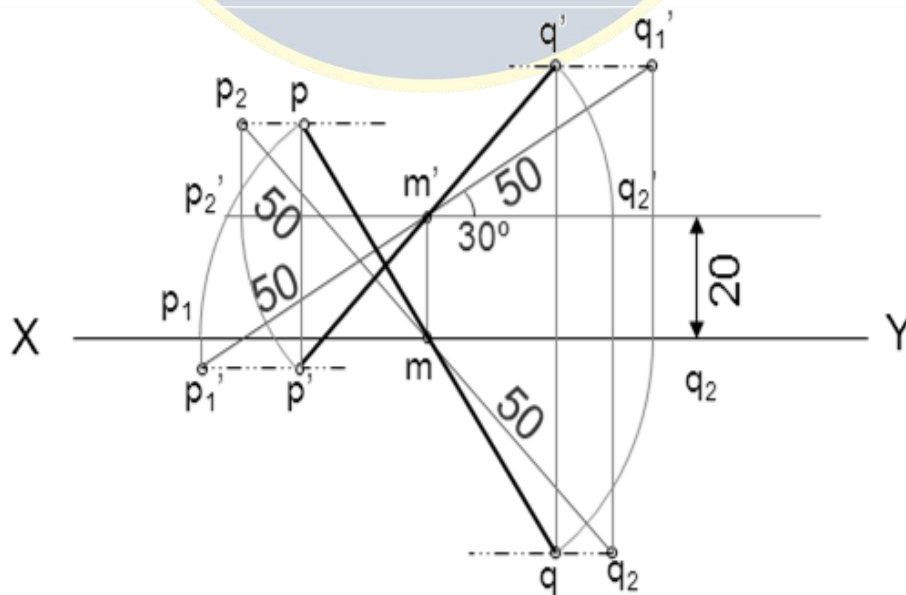


Fig.2.20

- a. Three different reference planes and their respective views Horizontal Plane, Vertical Plane and Side or Profile Plane Front view is a view projected on VP Top View is a view projected on HP and Side View is a view projected on PP.
- b. A line when parallel to both the planes HP and VP, then the line has true length in both the front and top views.
- c. If the line is inclined only to HP, the Front view is a line having the true length (TL) and true inclination θ
- d. If the line is inclined only to VP the Top view is a line having the true length (TL) and true inclination ϕ
- e. First angle projections method the objects are placed in 1st Quadrant (FV above x-y line and TV below x-y line) which is above HP and in front of VP.
- f. Third angle projections method the objects are placed in 3rd Quadrant (FV below x-y line and TV above x-y line) which is below HP and behind VP.

2.3.1 Important Questions

1. A line PS 65mm has its end P 15mm above the HP and 15mm in front of the VP. It is inclined at 55 deg to the HP and 35 deg to the VP. Draw its projections.
2. A line CD, inclined at 25deg to the HP, measures 80mm in top view. The end C is in the first quadrant and 25mm and 15mm from the HP and the VP, respectively. The end D is at equal distance from the both the reference planes. Draw the projections, find true length, and true inclination with the VP.
3. A straight line ST has its end S, 10mm in front of the VP and nearer to it. The mid-point M line is 50mm in front of the VP and 40mm above HP. The front and top view measure 90mm and 120mm, respectively. Draw the projection of the line. Also find its true length and true inclinations with the HP and VP.
4. A line PQ has its end P, 10mm above the HP and 20mm in front of the VP. The end Q is 85mm in front of the VP. The front view of the line measures 75mm. the distance between the end projectors is 50mm. Draw the projections of the line and find its true length and its true inclinations with the VP and HP.

5. A line PF, 65mm has its end P, 15mm above the HP and 15mm in front of the VP. It is inclined at 55deg to the VP. Draw its projections.

2.3.2 Previous Paper Questions:

1. A line CD 60mm long has its end 'C' in both H.P and V.P. It is inclined at 30 to V.P. Draw the projections to H.P and 450.
 2. A point C is 40mm below H.P and 20mm behind V.P, another points D and E are 60mm above H.P and in front of V.P, 90mm below H.P and 45mm in front of V.P, respectively. Draw the projections of all points on same reference line.
 3. The end P of a straight line PQ is 20 mm above the H.P. and 30 mm in front of V.P. The end Q is 15 mm below the H.P. and 45mm behind the V.P. If the end projectors are 50 mm apart, Draw the projection of PQ and determine the true length, traces, and inclination with the reference planes.
 4. The front view of line inclined at 30° to V.P is 65mm long. Draw the projections of a line, when it is parallel to and 40mm above H.P. and one end being 20mm in front of V.P.
 5. A line PQ, 64 mm long has one of its extremities 20 mm in front VP and the other 50 mm above HP. The line is inclined at 40° to HP and 25° to VP. Draw its top and front view.
 6. The projection of a line AB has 35° inclinations in top view and 40° inclinations in the front view with an elevation length of 60 mm. If the end A is 10 mm below HP and B is 12 mm behind VP, Draw the projections and locate the traces keeping the line in the third quadrant.
- Line PQ has 72 mm length in the front view and 66 mm length in the top view. The end P is 48 mm below HP and 40 mm behind VP, while the end Q is 12 mm below HP. Draw the projection of the line, locate the traces, and determine the true length and inclinations of the line with the reference planes.

8. Assignments

S.No	Question	BL	CO
1	Draw the projection of following points which is lying on the reference line 25mm apart. i. A point A is 20mm above HP and 30mm in front of VP. ii. A point B is 20mm above HP and 30mm behind VP. iii. A point C is 20mm below HP and 30mm behind VP. iv. A point D is 20mm below HP and 30mm in front of VP. v. A point E is on HP and 30mm in front of VP.	1	2
2	A Line AB of 70mm long is inclined at 30° to HP and 45° to VP. The end A of the line is 15 above HP and 20 Infront of VP. Draw the Projections of the line.	1	2
3	A line PQ of 65mm long has its end P, 25mm above HP and 20mm in front of V.P. The end Q is 40mm above HP and 50mm in front of VP Draw its projections and find its inclinations with HP and VP.	1	2
4	A hexagonal plane of side 30mm has an edge on the HP. Its surface is inclined at 45° to HP and the edge on which the plane rests is inclined at 30° to VP. Draw its projections	1	2
5	A semi – circular plate of 80mm diameter, has its straight edge on VP and inclined at 30° to HP, while the surface of the plate is inclined at 45° to V.P. Draw the projections of the plate.	1	2

9. Part B- Questions

S.No	Question	BL	CO
1	A cube of 35mm long edges is resting on HP on one of its faces with a vertical face inclined at 30° to VP. It is cut by a section plane parallel to VP and 9mm away from the axis and further away from VP. Draw its sectional front view and top view	1	2
2	A triangular pyramid having base 40mm side and axis 50mm long is lying on HP on one of its faces with the axis parallel to VP. A section plane parallel to VP cuts the pyramid at a distance of 6mm from the axis. Draw its front view and sectional top view.	1	2
3	A square pyramid base 40mm side and axis 65mm long is lying on based on HP and all the edges of the base are equally inclined to VP. It is cut by a section plane perpendicular to VP and inclined at 45° to HP and bisecting the axis. Draw its sectional top view and true shape of the section and sectional side view.	1	2
4	A cylinder of 40mm in diameter and 60mm height and having its axis vertical is cut by a section plane perpendicular to VP, inclined at 45° to HP and intersecting the axis 32mm above the base. Draw its front view, sectional top view, and true shape of the section.	1	2
5	A cone is resting on HP with its base of diameter 50mm and axis 70mm and it is cut by a section plane parallel to the base at a height of 20mm from the base. Draw its sectional top view.	1	2

10. Supportive Online Certification Courses

1. Engineering drawing and computer graphics By Prof. Rajaram Lakkaraju, conducted by IIT Kharagpur – 12 weeks

11. Prescribed Textbooks & Reference Books

Textbook

1. K.L. Narayana & P. Kannaiah, Engineering Drawing, 3/e, SciTech Publishers, Chennai, 2012.
2. N.D. Bhatt, Engineering Drawing, 53/e, Charotar Publishers, 2016.

References:

1. Venugopal, Engineering Drawing and Graphics, 3/e, New Age Publishers, 2000.
2. K.C. John, Engineering Graphics, 2/e, PHI, 2013.

